

Using ggplot2

EVR 628- Intro to Environmental Data Science

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Table of contents

Part 1: Build a basic plot	2
The <code>ggplot</code> function	3
The <code>aes</code> function	3
The <code>geom_point</code> function	4
<code>labs</code> function	5
Part 2: Visualizing distributions	6
Distribution of a cateogrical variable	6
Distribution of a numeric variable	7
Part 3: Visualizing relationships	8
A numerical and a categorical variable	8
Two categorical variables	9
Three or more	10
Part 4: Modifying colors	11
<code>scale_*_viridis_*</code>	11
<code>viridis</code> options	12
<code>scale_color_gradientn</code>	13
<code>scale_color_gradient2</code>	14
Helpful resources	20
Part 5: Extra material if time allows	20
Saving plots	20
Layering	21
Facetting	22

Part 1: Build a basic plot

- Open RStudio
- Make sure you are in the EVR628 project
- Create a new script, and save it to scripts as `week3_ggplot.R`
- Add a basic descriptive header
- Load packages

```
library(EVR628tools)
```

v v v You successfully loaded the EVR628tools package v v v

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

- Load data

```
data(data_lionfish)
```

- Inspect data

```
glimpse(data_lionfish)
```

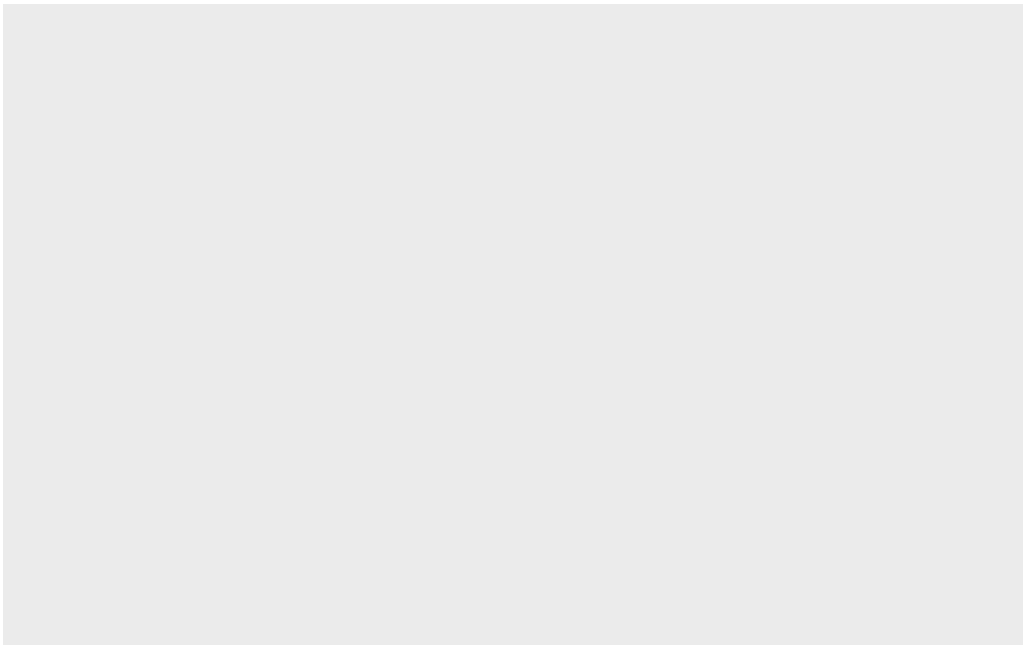
```
Rows: 109
Columns: 9
$ id          <chr> "001-Po-16/05/10", "002-Po-29/05/10", "003-Pd-29/05/10~
$ site        <chr> "Paraiso", "Paraiso", "Pared", "Canones", "Canones", "~
$ lat         <dbl> 20.48361, 20.48361, 20.50167, 20.47694, 20.47694, 20.5~
$ lon         <dbl> -87.22611, -87.22611, -87.21167, -87.23278, -87.23278,~
$ total_length_mm <dbl> 213, 124, 166, 203, 212, 210, 132, 122, 224, 117, 211,~
```

```
$ total_weight_gr <dbl> 112.70, 27.60, 52.30, 123.10, 129.00, 138.75, 50.29, 1~  
$ size_class      <chr> "large", "medium", "medium", "large", "large", "large"~  
$ depth_m         <dbl> 38.1, 27.9, 18.5, 15.5, 15.0, 22.7, 13.4, 18.5, 18.2, ~  
$ temperature_C   <dbl> 28, 28, 28, 28, 28, 29, 29, 29, 29, 29, 28, 28, 28, 28~
```

The ggplot function

- Show documentation: point out data and mapping
- Specify data

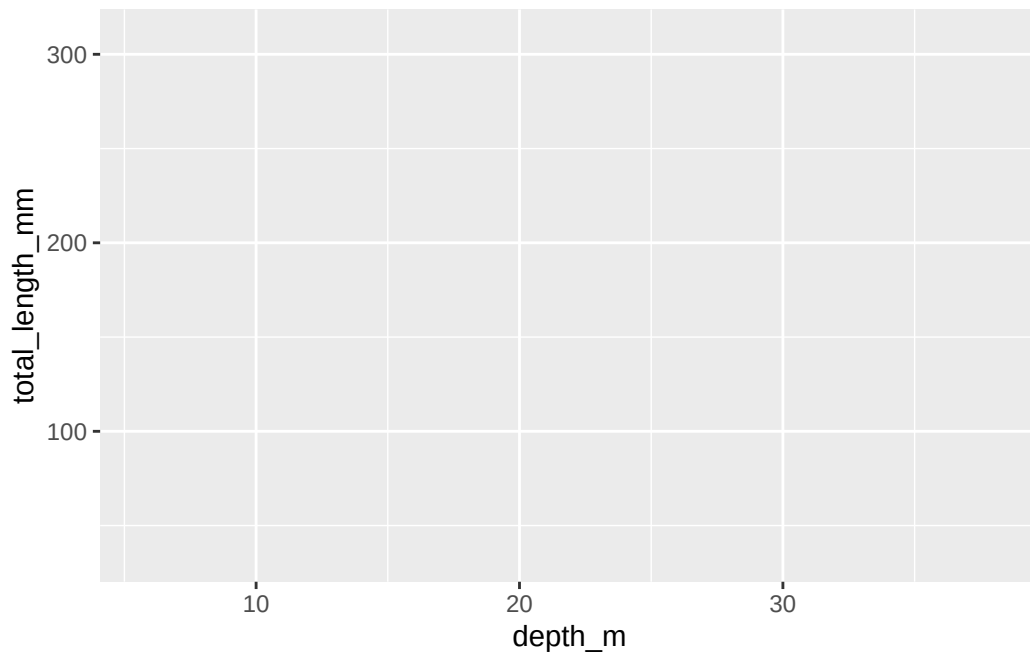
```
ggplot(data = data_lionfish)
```



The aes function

- Show documentation
- Explain a function inside a function
- Then specify mapping
 - Emphasis on use of commas and named arguments without quotations
- Multiple lines and parentheses

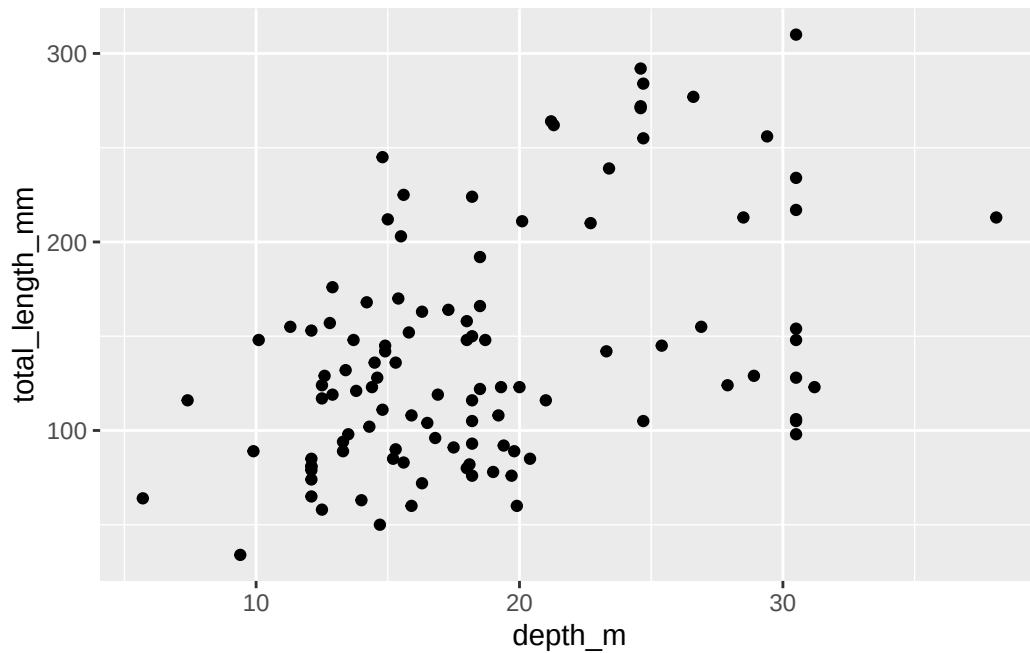
```
ggplot(data = data_lionfish,  
       mapping = aes(x = depth_m, y = total_length_mm))
```



The `geom_point` function

- Emphasis on + for layering
- Run with simple points
- Show documentation and scroll down to the **Aesthetics** section
- Show how to change colors / sizes etc...
- Emphasize difference between using variables and constant values (inside vs outside aes)

```
ggplot(data = data_lionfish,  
       mapping = aes(x = depth_m, y = total_length_mm)) +  
  geom_point()
```

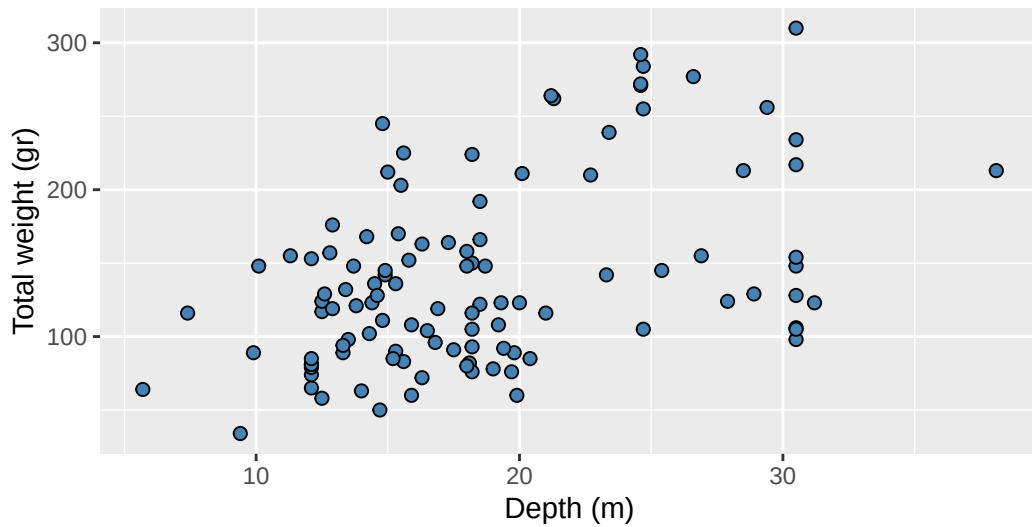


labs function

```
ggplot(data = data_lionfish,  
       mapping = aes(x = depth_m,  
                     y = total_length_mm)) +  
geom_point(shape = 21,  
           fill = "steelblue",  
           size = 2) +  
labs(x = "Depth (m)",  
     y = "Total weight (gr)",  
     title = "Body length and depth",  
     subtitle = "Larger fish tend to live deeper",  
     caption = "Source EVR628tools::data_lionfish")
```

Body length and depth

Larger fish tend to live deeper



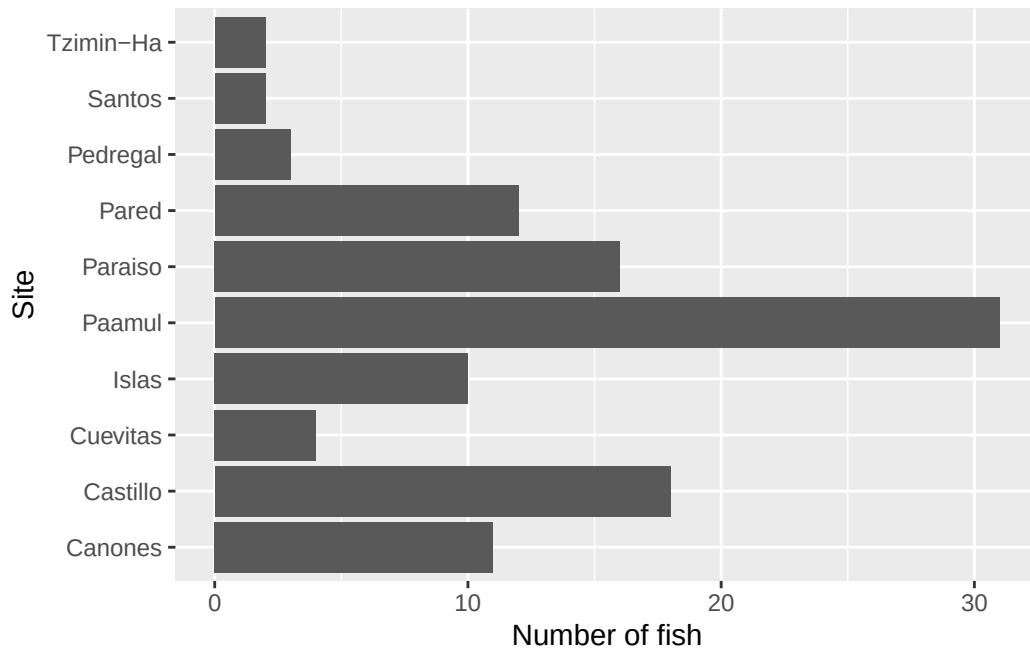
Source EVR628tools::data_lionfish

Part 2: Visualizing distributions

Distribution of a categorical variable

- Categorical variable with `geom_bar()`
 - basic with site

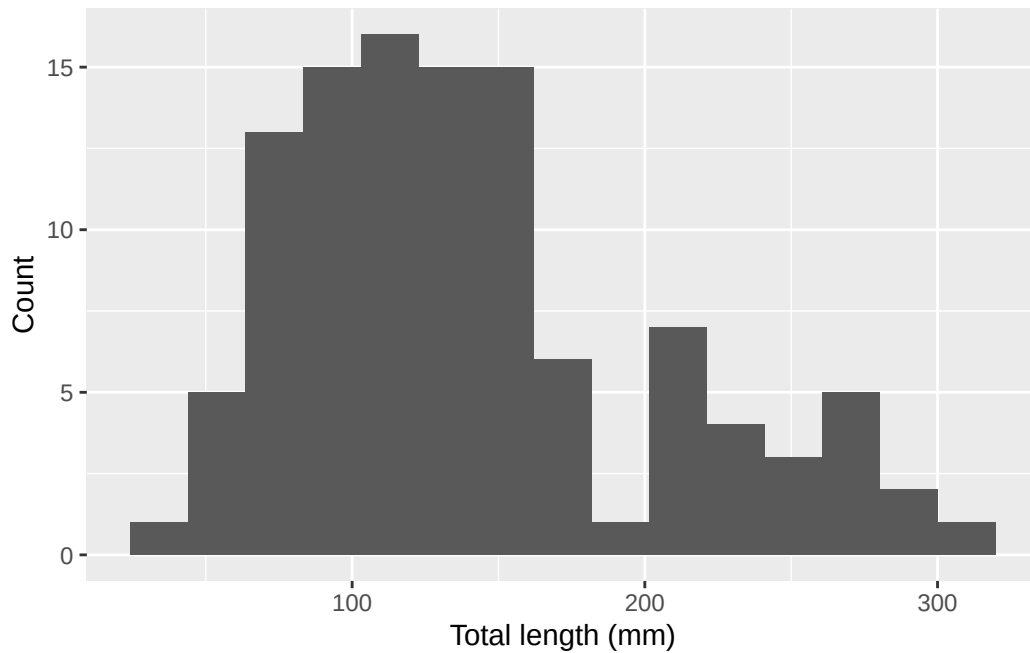
```
# Basic bar plot with site
ggplot(data = data_lionfish, aes(x = site)) +
  geom_bar() +
  coord_flip() +
  labs(x = "Site", y = "Number of fish")
```



Distribution of a numeric variable

- Numerical variable with `geom_histogram()`
 - Distribution of lengths
 - Show bins vs bin width
 - $k = 1 + 3.322 \times \log_{10}(N)$

```
# Histogram of lengths
ggplot(data = data_lionfish, aes(x = total_length_mm)) +
  geom_histogram(bins = 15) +
  labs(x = "Total length (mm)", y = "Count")
```

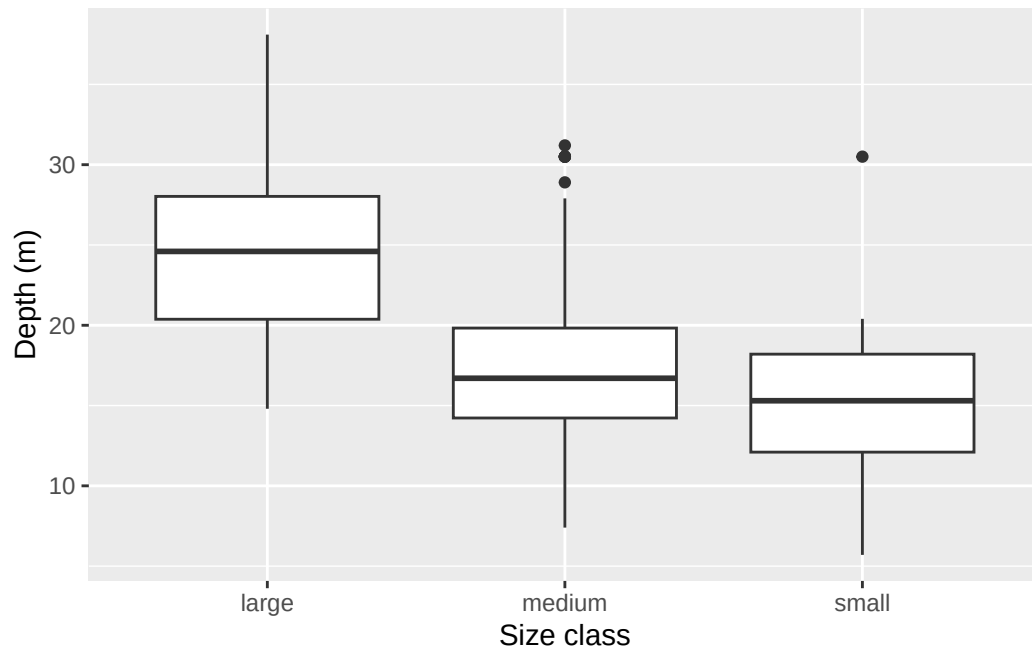


Part 3: Visualizing relationships

A numerical and a categorical variable

- depth by size class

```
# Numerical vs categorical: depth by size class
ggplot(data = data_lionfish, aes(x = size_class, y = depth_m)) +
  geom_boxplot() +
  labs(x = "Size class", y = "Depth (m)")
```

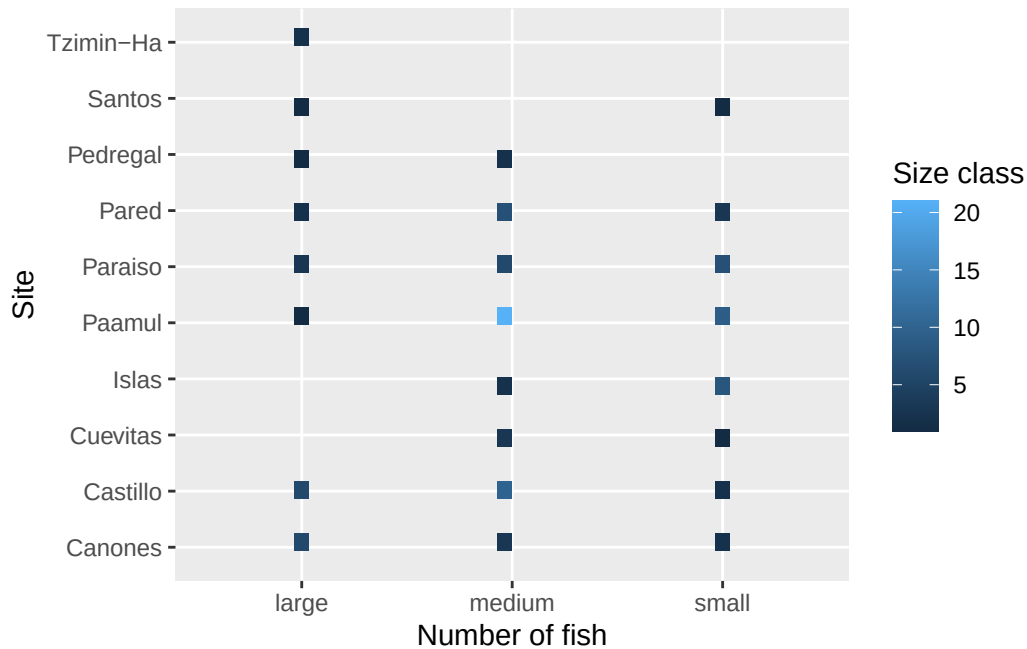



Two categorical variables

- N by size class and site

```
# Two categorical variables: N by size class and site
ggplot(data = data_lionfish, aes(x = site, y = size_class)) +
  geom_bin2d() +
  coord_flip() +
  labs(x = "Site", y = "Number of fish", fill = "Size class")
```

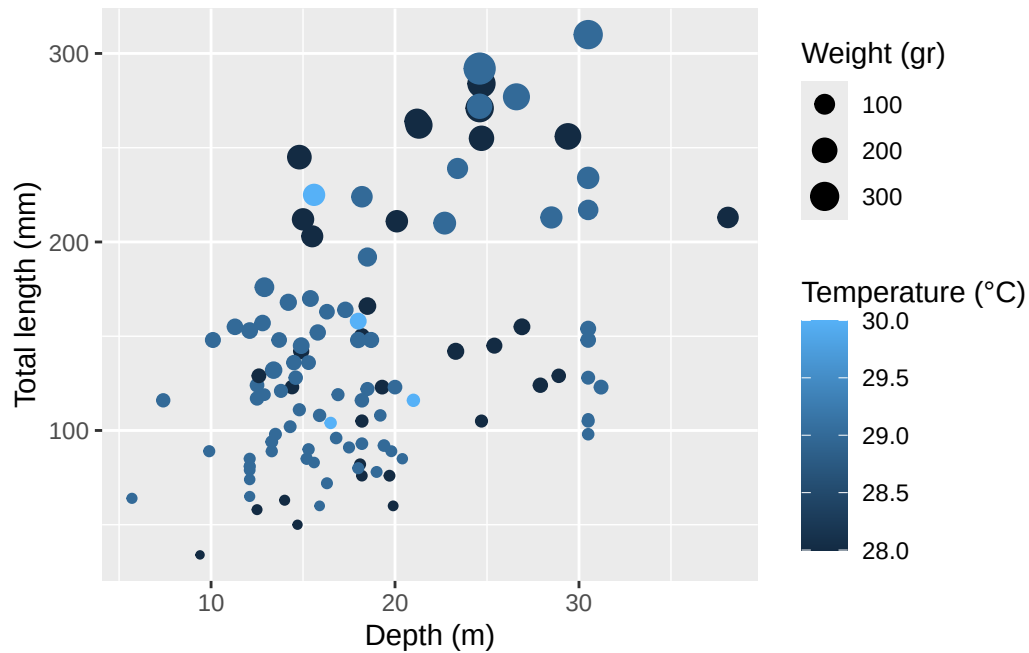
``stat_bin2d()`` using ``bins = 30``. Pick better value ``binwidth``.



Three or more

- length vs depth with size for weight and color for temperature

```
# Three or more variables: length vs depth with size for weight
ggplot(data = data_lionfish,
       aes(x = depth_m, y = total_length_mm,
           size = total_weight_gr,
           color = temperature_C)) +
geom_point() +
scale_size_continuous(range = c(1, 5)) +
labs(x = "Depth (m)", y = "Total length (mm)",
     size = "Weight (gr)", color = "Temperature (°C)")
```



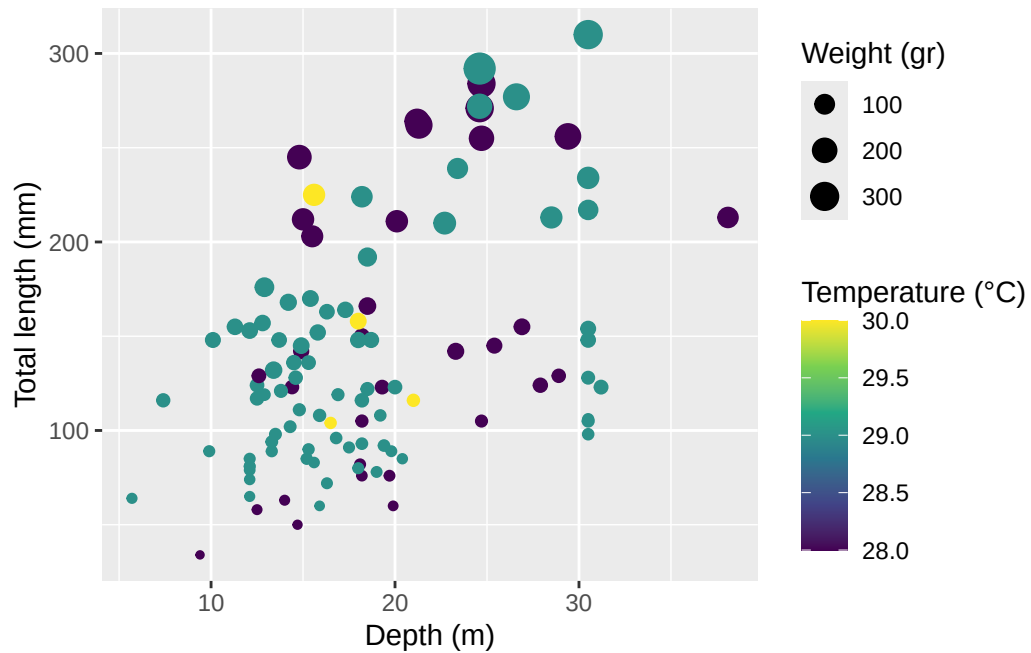
Part 4: Modifying colors

scale*_viridis_*

- The easiest way is to use `scale*_viridis_*`
 - Check documentation
 - An example

```
p <- ggplot(data = data_lionfish,
  aes(x = depth_m, y = total_length_mm,
    size = total_weight_gr,
    color = temperature_C)) +
  geom_point() +
  scale_size_continuous(range = c(1, 5)) +
  labs(x = "Depth (m)", y = "Total length (mm)",
    size = "Weight (gr)", color = "Temperature (°C)") +
  scale_color_viridis_c()
```

p



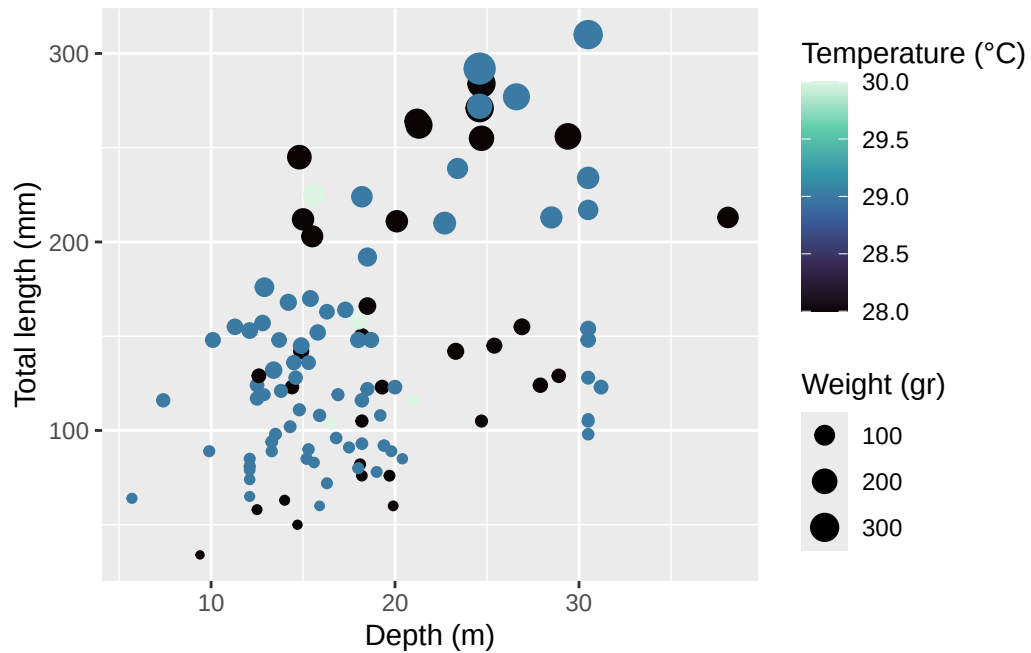
viridis options

- Show errors for wrong type of scale
- Change to mako

```
p +
  scale_color_viridis_c(option = "mako")
```

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.



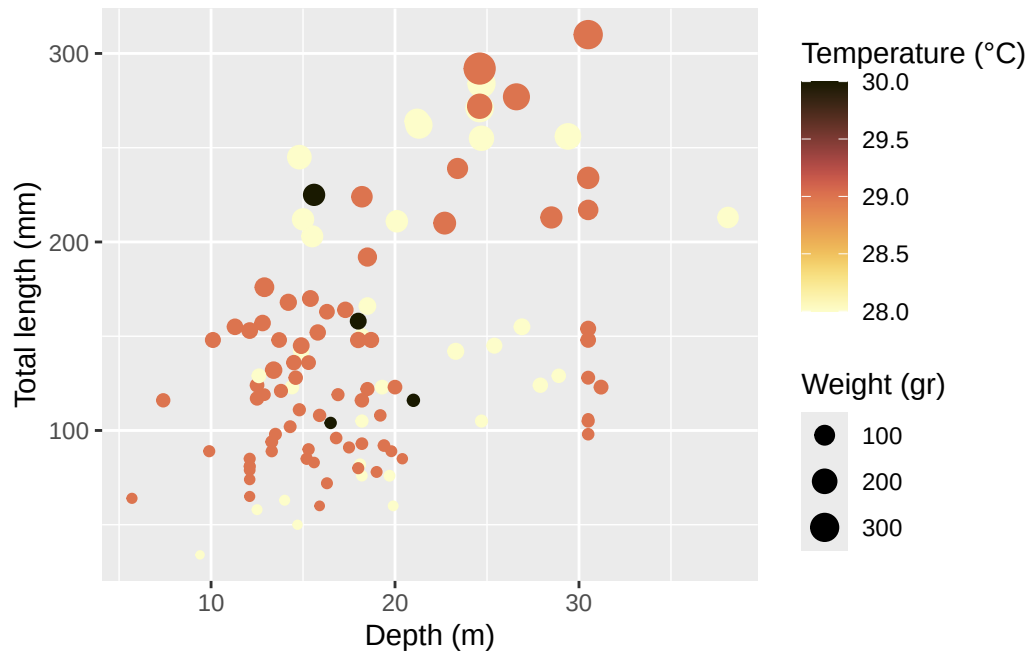
scale_color_gradientn

- `scale_color_gradientn` for continuous and custom palette
 - Check documentation
 - An example

```
p +
  scale_color_gradientn(colours = palette_IPCC(var = "temp", type = "seq"))
```

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.



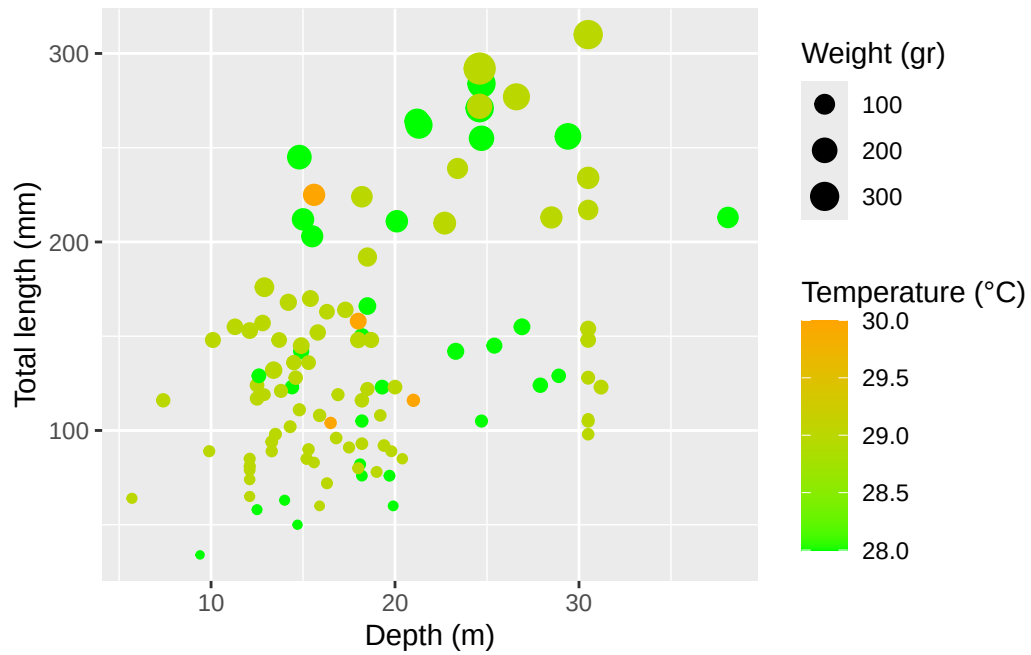
scale_color_gradient2

- `scale_color_gradient` vs `scale_color_gradient2` for continuous / divergent with ends
 - Check documentation
 - An example

```
# Green to orange
p +
  scale_color_gradient(low = "green",
                      high = "orange")
```

Scale for colour is already present.

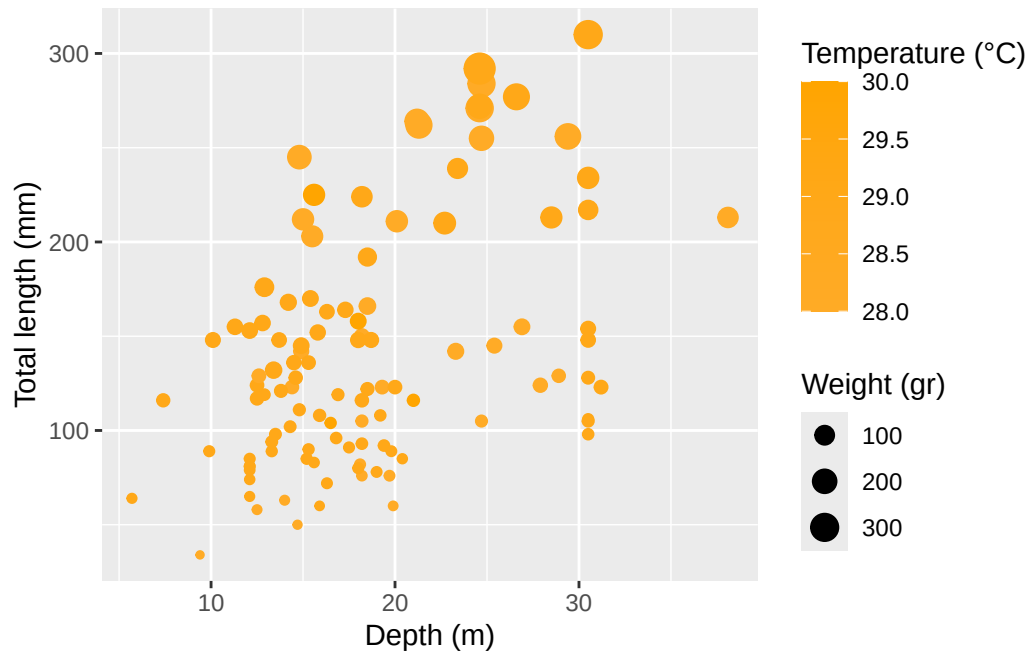
Adding another scale for colour, which will replace the existing scale.



```
# Going through white
p +
  scale_color_gradient2(low = "green",
                        mid = "white",
                        high = "orange")
```

Scale for colour is already present.

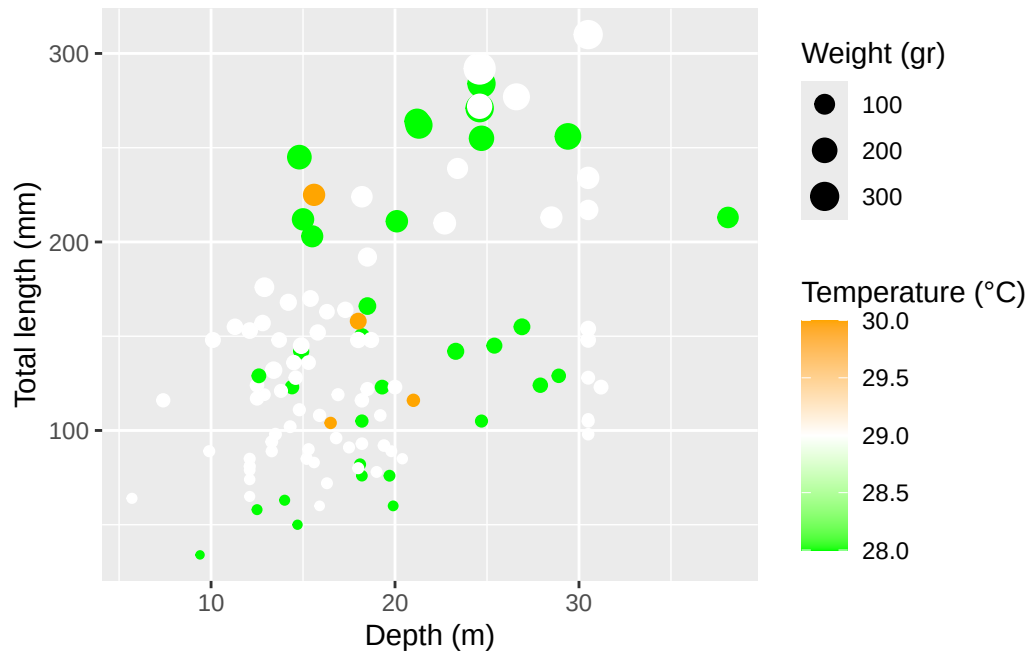
Adding another scale for colour, which will replace the existing scale.



```
# Set white at 0
p +
  scale_color_gradient2(low = "green",
                        mid = "white",
                        high = "orange",
                        midpoint = 29)
```

Scale for colour is already present.

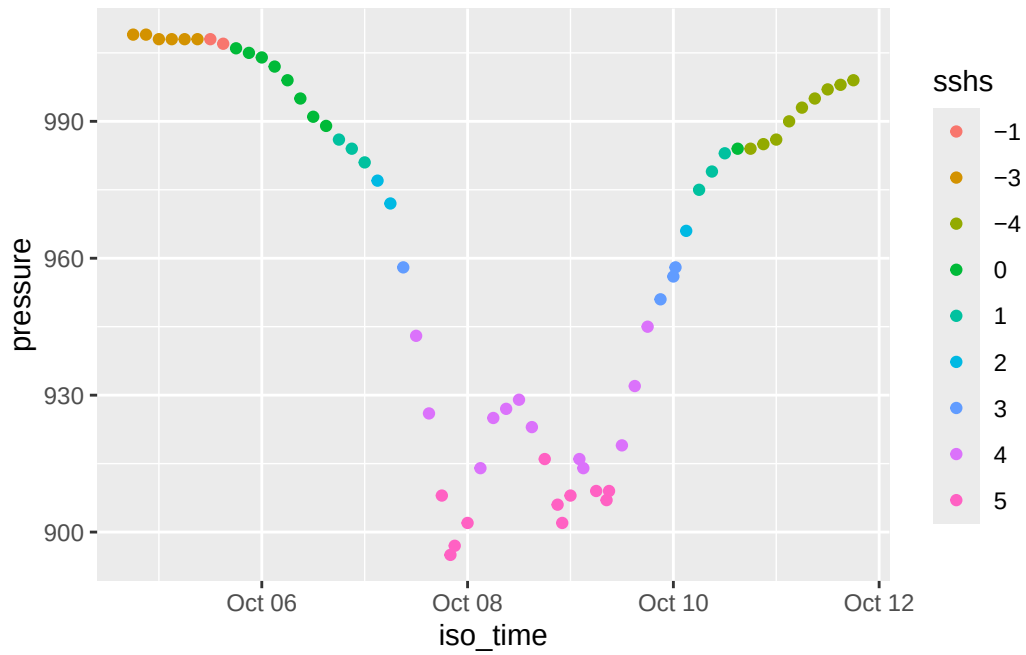
Adding another scale for colour, which will replace the existing scale.



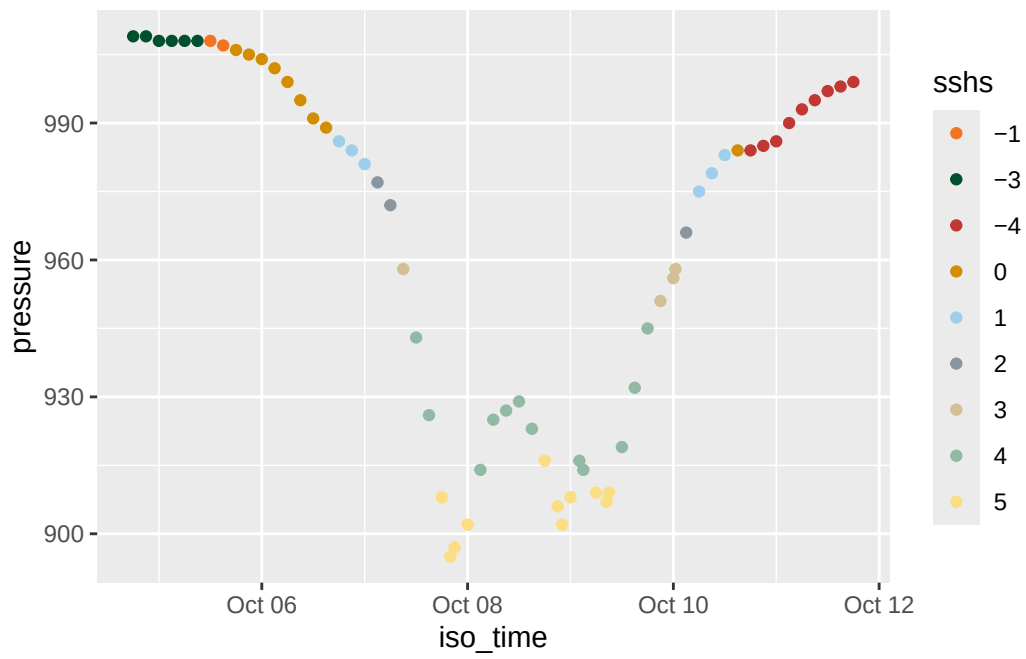
- `scale_color_manual` for discrete
 - Check documentation
 - An example

```
p <- ggplot(data = data_milton,
            aes(x = iso_time, y = pressure, color = sshs)) +
  geom_point()

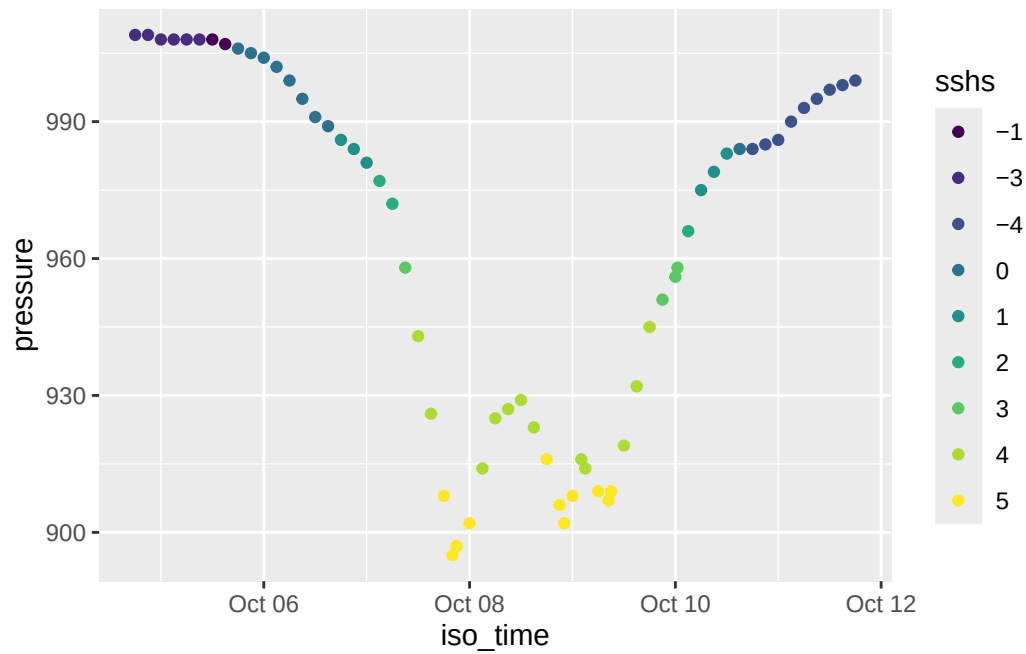
p
```



```
p +  
  scale_color_manual(values = palette_UM(n = 10))
```

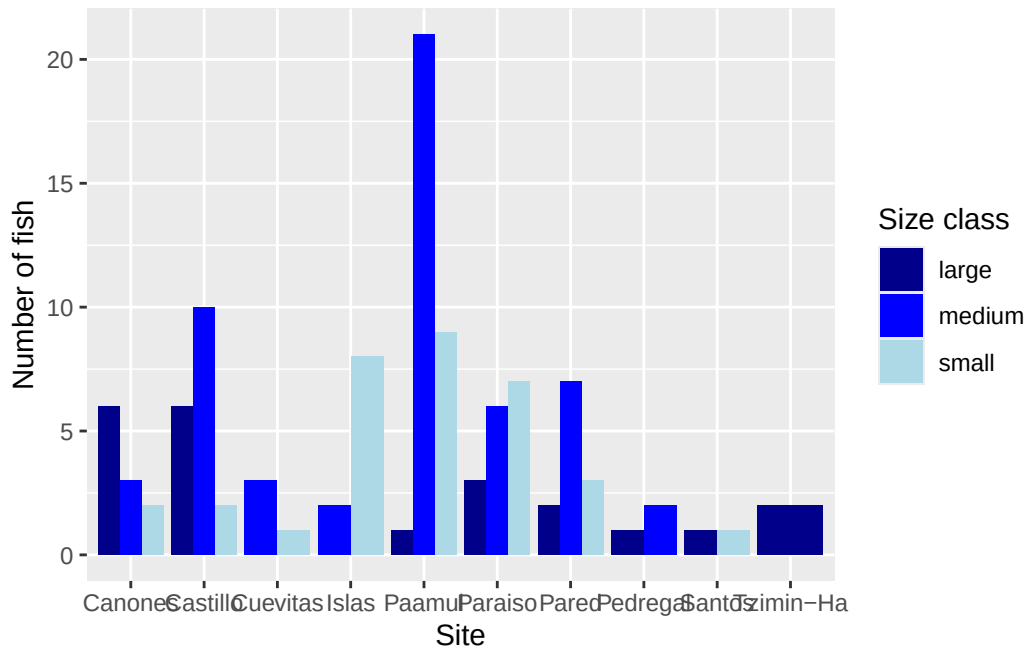


```
p +  
  scale_color_viridis_d()
```



- `scale*_manual`

```
# Example with manual color scale for discrete variables  
ggplot(data = data_lionfish, aes(x = site, fill = size_class)) +  
  geom_bar(position = "dodge") +  
  scale_fill_manual(values = c("small" = "lightblue",  
                                "medium" = "blue",  
                                "large" = "darkblue")) +  
  labs(x = "Site", y = "Number of fish", fill = "Size class")
```



Helpful resources

- [ggplot cheatsheet](#)

Part 5: Extra material if time allows

Saving plots

- Repeat the entire process, one line at a time, showing how to save to an object
- Then show how to save plot with ggsave
- Show documentation of ggsave
- Emphasize creating plot objects

```
# Save plot to object
p <- ggplot(data = data_lionfish,
            mapping = aes(x = depth_m, y = total_length_mm)) +
  geom_point(shape = 21, fill = "steelblue", size = 2) +
  labs(x = "Depth (m)",
       y = "Total length (mm)",
       title = "Body length and depth")
```

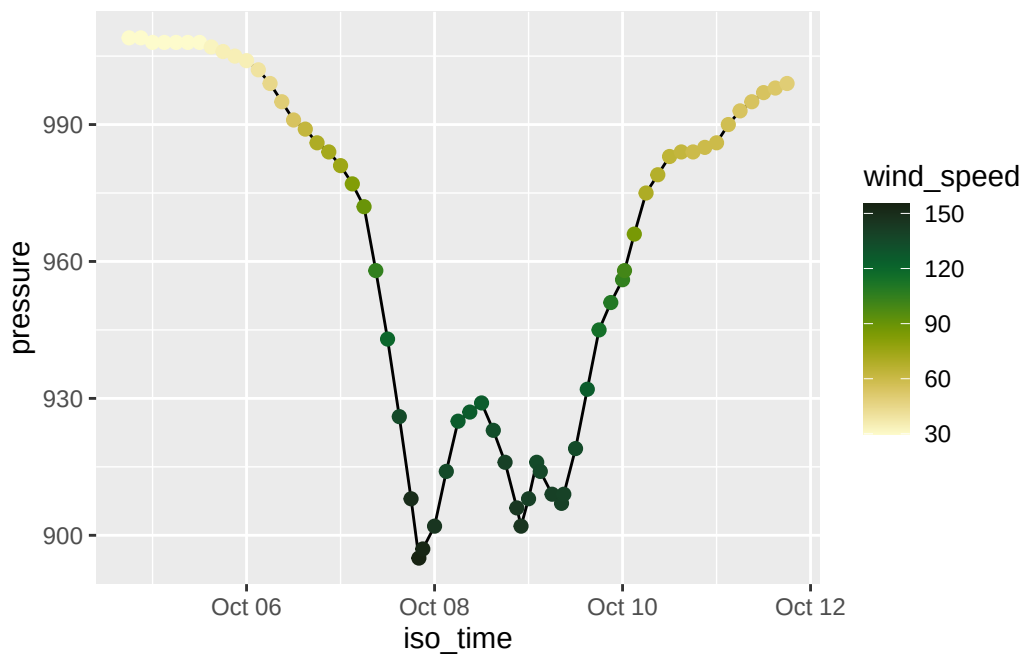
```
# Save plot to file
ggsave("lionfish_depth_length.png", p, width = 8, height = 6, dpi = 300)
```

Layering

- Show `data_milton` documentation again
- Build plot below
- Show use of `color` vs `fill` inside main `aes()` (affecting line and points) vs `aes()` inside `geom_point()` (affecting only the points)

```
p <- ggplot(data = data_milton,
            mapping = aes(x = iso_time,
                          y = pressure,
                          color = wind_speed,
                          size = 2)
            ) +
  geom_line() +
  geom_point(aes(color = wind_speed),
            size = 2)

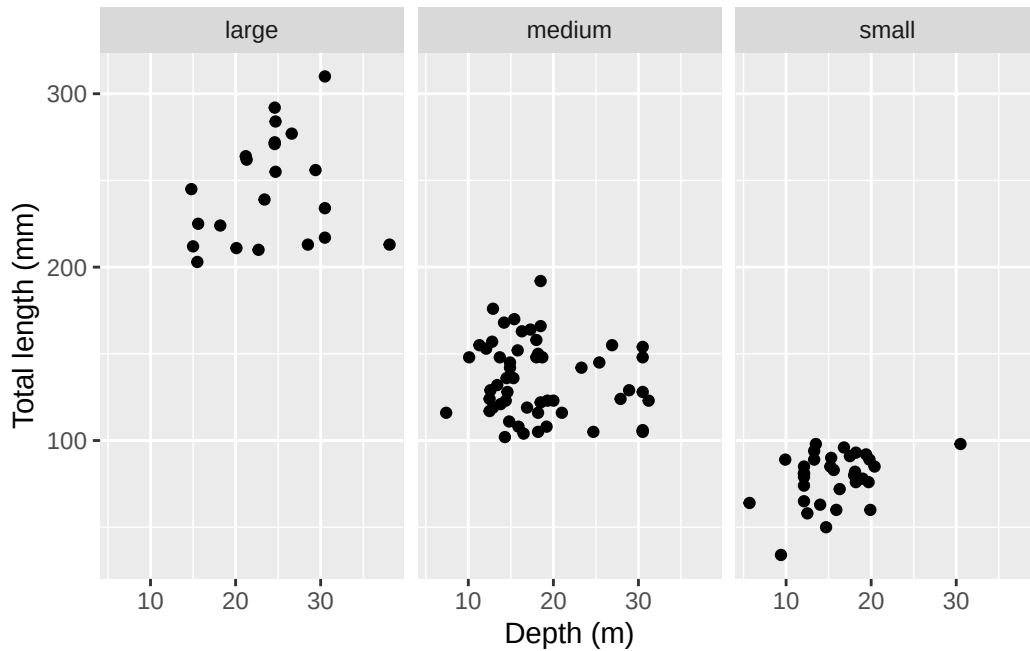
p +
  scale_color_gradientn(colours = palette_IPCC(var = "wind", type = "seq"))
```



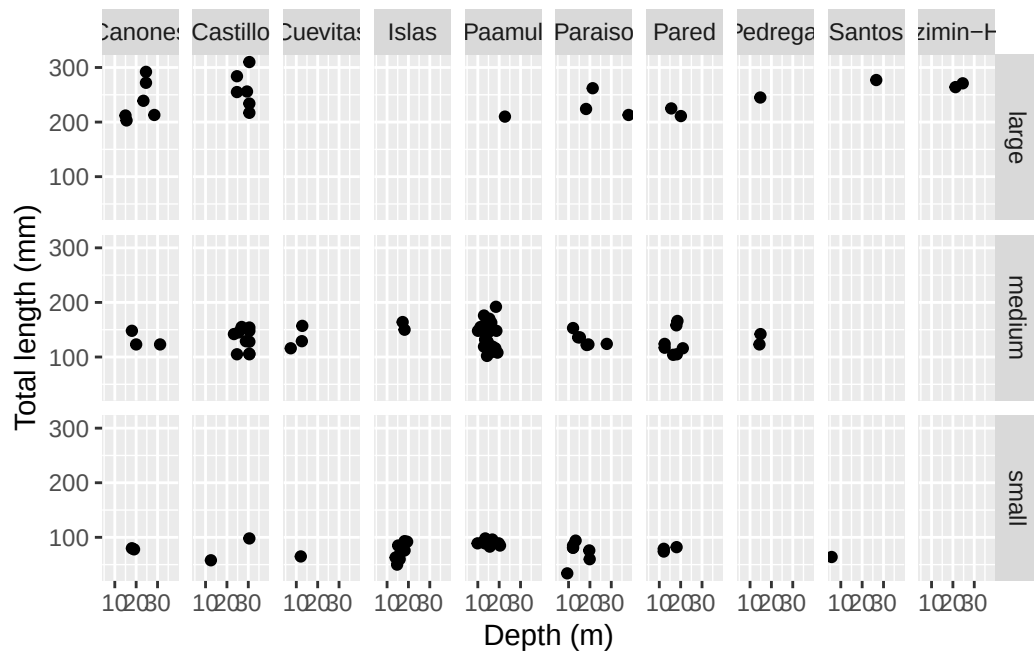
Facetting

- Show documentation for `facet_wrap` and `facet_grid`
- Show how each of these work

```
# Facet wrap example
ggplot(data = data_lionfish, aes(x = depth_m, y = total_length_mm)) +
  geom_point() +
  facet_wrap(~size_class, ncol = 3) +
  labs(x = "Depth (m)", y = "Total length (mm)")
```



```
# Facet grid example
ggplot(data = data_lionfish, aes(x = depth_m, y = total_length_mm)) +
  geom_point() +
  facet_grid(size_class ~ site) +
  labs(x = "Depth (m)", y = "Total length (mm)")
```



```
# Facet wrap with different scales
ggplot(data = data_lionfish, aes(x = depth_m, y = total_length_mm)) +
  geom_point() +
  facet_wrap(~size_class, ncol = 3, scales = "free_y") +
  labs(x = "Depth (m)", y = "Total length (mm)")
```

